

Technical Design: Multi-Source Candidate Data Transformer

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1. Pipeline Steps & Architecture

Pipeline:

[Ingestion] → [Parsing] → [Normalization] → [Merge & Deduplication] → [Projection] → [Validation]

Ingestion: Detect source type (CSV, JSON, PDF, TXT).

Parsing: Extract structured and unstructured candidate fields using parsers and regex.

Normalization: Standardize phones, dates, countries, and skills.

Merge: Deduplicate and merge candidate records using email/phone matching.

Confidence & Provenance: Track source, extraction method, and confidence score.

Projection & Validation: Apply runtime config and validate final schema.

2. Canonical Schema & Normalization

Canonical schema contains:

candidate_id, full_name, emails, phones, location, links, headline, skills, education, experience, provenance, overall_confidence.

Normalization Rules:

- Phones → E.164 format (+919876543210)
- Dates → YYYY-MM format
- Countries → ISO-3166 alpha-2 codes
- Skills → canonicalized skill names (ReactJS → React)

3. Conflict Resolution & Confidence

Entity Matching: email, phone, or normalized name matching.

Trust Hierarchy: ATS JSON > Recruiter CSV > LinkedIn/GitHub > Resume > Notes.

Provenance: Every field stores source and extraction method.

Confidence Score: Calculated using field completeness, validation success, and source reliability.

4. Runtime Output Configuration

The pipeline supports config-driven output projection without modifying core logic.

Supported operations:

- Rename fields
- Select subset of fields
- Convert arrays to single values
- Enable/disable provenance

Example: emails[0] → primary_email

5. Edge Cases & Descoping

Handled Edge Cases:

- Missing email/phone
- Corrupted PDF text
- Duplicate skills
- Malformed CSV rows
- Partial candidate records

Descoped: Advanced NLP parsing, real-time DB sync, and full LinkedIn API integration due to time constraints.